

Department of Mathematical Sciences
Carnegie Mellon University

Summer 2021

MSCF Math Prep

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Lectures: 1:00 - 3:00 PM, August 3,4,5, 10, 11, 12, 17, 18, 19

Office Hours: There will be frequent office hours in person and online. We will take time zone considerations into account where appropriate.

Trips to NY: Prof. Hrusa will be in New York all day on Monday August 16 and Tuesday August 17.

Course Description: This is a “refresher” course in calculus (single and multi variable), linear algebra, and differential equations, with emphasis on aspects of those subjects that will be most important for success in the MSCF program and in Quant Interviews. (You can think of this as *Math Boot Camp*.) I will not focus on theory, except as it is needed to solve problems. I will not cover probability theory (because this will be covered in Probability Prep). However, I will go over the use of techniques from calculus to understand moment generating functions and I will do some calculations involving normal and log-normal distributions that are relevant to computing option prices. Lecture Notes, Problem Sets, etc. will be posted on Canvas.

Problem Sets: I plan to assign 6 problem sets They will be self-contained – all problems will be stated completely so that you will not have to refer to books. The problem sets do not need to be turned in. However, if you would like to get feedback on your written work, you may attach your solutions as a single PDF file to the assignment Canvas by 11:59 PM on the due date. (You can hand in solutions to selected problems if you want.)

Grading: The course will be graded Pass/Fail. The grade will be based completely on the Final Exam. The passing score is 80%. (More info on the final exam will be provided later in the course.) Students who do not pass the final on the first attempt can take a make-up final exam without any penalty. I am anticipating that virtually all students will pass the final exam on the first attempt. If there is anyone who fails both the original final and the make-up final, a decision on how to proceed will be made in consultation with the MSCF Steering Committee. (I regard this as an unlikely scenario – it would indicate that the student is probably not ready to start the MSCF Program.)

Final Exam: The final exam will be given on Tuesday August 24 from 9:00 AM to Noon. It will be closed-book and closed-notes. Calculators will **not** be permitted. The exam will be comprehensive (covering all material from the course).

Recommended References: There is no official textbook for the course. Here are several recommended references.

1. Your old calculus textbook.
2. *Calculus*, F. Ayres & E. Mendelson, Schaum's Outlines, McGraw-Hill, 5th Ed.
3. *Linear Algebra*, D. Poole, Cengage
4. *Elementary Differential Equations and Boundary Value Problems*, W. Boyce & R. DiPrima, Wiley
5. *A Primer for the Mathematics of Financial Engineering*, D. Stefanica, FE Press
(This book should be very useful throughout the MSCF Program.)